

# Supplementary Information

Resonant versus geometric anisotropy of the Purcell enhancement near a gold nanorod

## S1. Boundary-element calculations: validation and convergence

All boundary-element (BEM) results were obtained with MNPBEM17 [3–5] (retarded solver, curved boundary elements, Johnson–Christy gold from the tabulated file distributed with the toolbox) in MATLAB R2026b on GitHub-hosted runners. The script `isi/bem/run_mnpbem.m` defines all cases; the workflow `.github/workflows/bem-runs.yml` runs one case per job. Table S1 compares the equal-volume sphere with the exact Mie solution evaluated with the same tabulated permittivity (radial dipole: Ruppel [6]; tangential dipole: Kim *et al.* [7]); Table S2 and Fig. S1 give the mesh convergence of the rod.

Table S1: BEM versus exact Mie theory for the equal-volume gold sphere ( $R = 15.874$  nm). Median absolute relative deviation over 500 nm to 900 nm (maximum in brackets for the radial  $F_p$ ).

| mesh (vertices) | host  | gap (nm) | radial $F_p$ | radial $T$ | tangential $F_p$ | tangential $T$ |
|-----------------|-------|----------|--------------|------------|------------------|----------------|
| 400             | air   | 5        | 2.4% (6.5%)  | 1.8%       | 2.1%             | 1.0%           |
| 784             | air   | 5        | 1.0% (4.4%)  | 1.4%       | 0.7%             | 0.9%           |
| 1444            | air   | 5        | 0.9% (4.2%)  | 1.4%       | 0.7%             | 1.0%           |
| 1444            | air   | 10       | 1.5% (3.8%)  | 1.3%       | 0.5%             | 1.1%           |
| 1444            | air   | 20       | 0.9% (3.3%)  | 1.3%       | 0.1%             | 1.2%           |
| 784             | water | 5        | 0.6% (4.4%)  | 1.4%       | 0.7%             | 0.8%           |
| 784             | water | 20       | 0.4% (2.9%)  | 1.3%       | 0.1%             | 1.2%           |

Table S2: Mesh convergence of the rod (axial dipole).  $h$ : target element size.

| $h$ (nm) | elements | gap (nm) | $F_p^{\max}(\lambda)$ | $T_{\max}(\lambda)$ | $\eta_a(\lambda_T)$ |
|----------|----------|----------|-----------------------|---------------------|---------------------|
| 3        | 720      | 5        | 1649 (604)            | 117.7 (608)         | 7.3%                |
| 2        | 1550     | 5        | 1643 (604)            | 117.0 (608)         | 7.3%                |
| 1.5      | 2829     | 5        | 1638 (604)            | 116.5 (608)         | 7.3%                |
| 3        | 720      | 3        | 3672 (508)            | 209.8 (608)         | 6.5%                |
| 1.5      | 2829     | 3        | 3782 (508)            | 214.1 (608)         | 6.6%                |
| 1.5      | 2829     | 20       | 78 (604)              | 8.2 (612)           | 11.6%               |

## S2. Plasmon-eigenmode decomposition, excitation and additional set-ups

*Eigenmodes.* The 40 lowest eigenvalues of the quasistatic BEM matrix  $F$  were computed with `bemstateig` of MNPBEM17 on the 2 nm meshes (sphere: 784 vertices). The longitudinal (transverse) dipolar mode is the eigenvector  $u_k$  with the largest oscillator strength  $s_k = (\sum_j a_j \mathbf{r}_j u_{k,j}) \cdot$

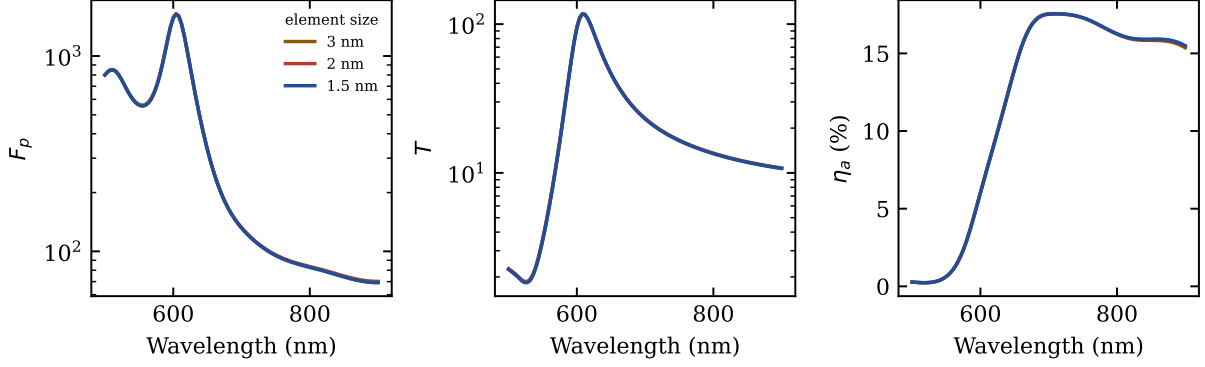


Figure S1: Mesh convergence of the BEM for the rod with axial dipole at a 5 nm gap.

$(\tilde{u}_k \cdot \mathbf{n})$  for a uniform field along  $z$  ( $x$ ), where  $a_j$ ,  $\mathbf{r}_j$  and  $\mathbf{n}_j$  are the area, centroid and normal of element  $j$ ; its contribution to  $F_p - 1$  follows from Eq. (3) of the main text and is evaluated with the same decay-rate routine as the full solution. The trivial eigenvalue  $-2\pi$  (uniform charge) does not couple. Script: `isi/bem/extra/job_modes.m`. Table S3 lists the decomposition at the resonance of the longitudinal mode.

Table S3: Quasistatic decomposition of the axial  $F_p$  at the resonance of the longitudinal dipolar mode ( $\lambda_L$ ). Share: fraction of  $F_p - 1$  carried by that mode; background:  $F_p - 1$  minus that contribution, plus one; geometric ratio: background divided by the transverse  $F_p$ ; total ratio: axial over transverse  $F_p$ .

| particle | $\lambda_L$ (nm) | gap (nm) | axial $F_p$ | share | background | geometric ratio | total ratio |
|----------|------------------|----------|-------------|-------|------------|-----------------|-------------|
| rod AR 2 | 540              | 5        | 859         | 57%   | 368        | 2.7             | 6           |
|          |                  | 10       | 154         | 80%   | 32         | 2.6             | 13          |
| rod AR 3 | 595              | 3        | 2936        | 80%   | 593        | 2.7             | 13          |
|          |                  | 5        | 1374        | 91%   | 129        | 3.1             | 33          |
|          |                  | 10       | 352         | 96%   | 14         | 3.3             | 82          |
|          |                  | 20       | 55          | 98%   | 1.9        | 1.6             | 47          |
| rod AR 4 | 655              | 5        | 3095        | 98%   | 57         | 3.4             | 185         |
|          |                  | 10       | 861         | 99%   | 7.5        | 3.3             | 382         |
| rod AR 5 | 725              | 5        | 4661        | 99%   | 46         | 3.7             | 374         |
|          |                  | 10       | 1322        | 100%  | 6.8        | 3.6             | 693         |
| sphere   | 510              | 5        | 761         | 9%    | 693        | 2.4             | 3           |
|          |                  | 10       | 89          | 21%   | 70         | 2.4             | 3           |

*Substrate.* The rod (axis along  $x$ ) lies 1 nm above a glass half-space ( $\varepsilon = 1.52^2$ ); the reflected Green function is tabulated with `compgreentablayer` at 23 wavelengths between 480 and 920 nm and interpolated. Rates are given in units of the vacuum rate; the reference is the same dipole above bare glass (`decayrate0`). Script: `job_substrate.m`.

*Nonlocal response.* Following Luo *et al.* [8] the metal is shrunk by  $\Delta d = 0.05$  nm and covered by a layer of the same thickness with permittivity  $\varepsilon_t = \varepsilon_m \varepsilon_b q_L \Delta d / (\varepsilon_m - \varepsilon_b)$ ,  $q_L = \sqrt{\omega_p^2 / \varepsilon_\infty - \omega(\omega + i\gamma)} / \beta$ , with the tabulated Johnson–Christy  $\varepsilon_m$ ,  $\hbar\omega_p = 9.02$  eV,  $\varepsilon_\infty = 9.84$ ,  $\hbar\gamma = 0.071$  eV and  $\beta = \sqrt{3/5} v_F$ ,  $v_F = 1.40 \times 10^6$  m/s; integration over neighbouring elements of the two surfaces uses polar refinement (`npol` = 20). The local reference uses the same mesh. Script: `job_nonlocal.m`.

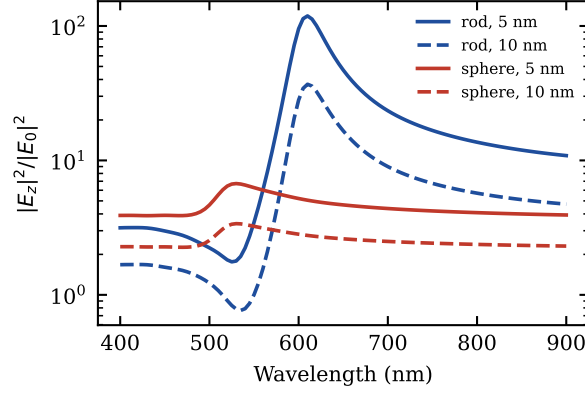


Figure S2: Local intensity enhancement  $|E_z|^2/|E_0|^2$  on the axis, 5 and 10 nm beyond the apex, for a plane wave polarised along the axis (BEM): rod and equal-volume sphere.

### S3. Drude–Lorentz representation of Johnson–Christy gold

For the time-domain body-of-revolution solver the Johnson–Christy permittivity [1] is represented as

$$\varepsilon(f) = \varepsilon_\infty - \frac{\sigma_0 f_0^2}{f^2 + i f \gamma_0} + \sum_{j=1}^2 \frac{\sigma_j f_j^2}{f_j^2 - f^2 - i f \gamma_j}, \quad (\text{S1})$$

with  $f = 1/\lambda$  in  $\mu\text{m}^{-1}$  (Meep convention,  $e^{-i\omega t}$ ). A least-squares fit over 440 nm to 1000 nm, weighted by  $|\varepsilon|^{-1/2}$ , gives the parameters of Table S4; the root-mean-square relative error is 1.6 % and the maximum 3.4 %. The same model is used in the Mie reference calculations, so that benchmark differences reflect discretisation only.

Table S4: Fit parameters ( $hcf$  gives energies in eV with  $hc = 1.2398 \text{ eV } \mu\text{m}$ ).

| term                 | $f_j$ ( $\mu\text{m}^{-1}$ ) | $hcf_j$ (eV) | $\gamma_j$ ( $\mu\text{m}^{-1}$ ) | $\sigma_j$ |
|----------------------|------------------------------|--------------|-----------------------------------|------------|
| $\varepsilon_\infty$ |                              | 2.229        |                                   |            |
| Drude                | 8.276                        | 10.26        | 0.0490                            | 0.701      |
| Lorentz 1            | 2.208                        | 2.737        | 0.509                             | 1.156      |
| Lorentz 2            | 2.755                        | 3.416        | 0.010                             | 2.101      |

### S4. Three-dimensional FDTD set-up

Lumerical FDTD 2024 R1, cubic domain of side 600 nm with 12 PML layers, maximum simulation time 300 fs, auto-shutoff  $10^{-5}$  (reached after 19 fs to 28 fs for cases B, C, D and 52 fs to 80 fs for the axial rod cases). A uniform override mesh ( $\Delta = 2 \text{ nm}$  unless stated) covers the particle, gap and source; global mesh accuracy 3. Radiated power is the flux through six planar monitors on a 400 nm cube; 201 wavelengths between 500 and 900 nm. The dipole sits at  $z = 35 \text{ nm}$ , between two nodes of the 2 nm mesh, whereas in the free-space reference (case B) it sits at  $z = 0$  on a node. The script that builds and runs all cases from scratch is provided with the data.

Table S5: All three-dimensional FDTD cases (raw definitions).  $\eta_{F_p}$ ,  $\eta_T$ : efficiency at the Purcell and radiative peaks;  $\eta_{\max}$ : spectral maximum.

| case                            | $\lambda_{F_p}$ | $F_p^{\max}$ | $\lambda_T$ | $T_{\max}$ | $\eta_{F_p}$ | $\eta_T$ | $\eta_{\max}$ |
|---------------------------------|-----------------|--------------|-------------|------------|--------------|----------|---------------|
| A rod, axial, 5 nm              | 610             | 1496         | 616         | 119        | 7.6%         | 8.6%     | 19%           |
| C rod, transverse               | 508             | 239          | 500         | 0.48       | 0.18%        | 0.21%    | 5.7%          |
| D sphere, radial                | 508             | 659          | 532         | 6.03       | 0.76%        | 1.1%     | 12%           |
| D $_{\perp}$ sphere, tangential | 510             | 268          | 500         | 0.45       | 0.15%        | 0.18%    | 1.9%          |
| D Mie (exact)                   | 506             | 772          | 530         | 6.58       | 0.67%        | —        | 11%           |
| G3 gap 3 nm, 1-nm mesh          | 516             | 7198         | 614         | 252        | 0.05%        | 6.4%     | 7.7%          |
| G10 gap 10 nm                   | 610             | 553          | 614         | 46.8       | 8.1%         | 8.9%     | 31%           |
| G20 gap 20 nm                   | 610             | 91           | 618         | 9.9        | 9.6%         | 12%      | 74%           |
| M3 mesh 3 nm                    | 644             | 2027         | 648         | 184        | 8.7%         | 9.3%     | 18%           |
| M1.5 mesh 1.5 nm                | 540             | 3752         | 620         | 165        | 0.05%        | 7.0%     | 9.6%          |

## S5. Free-space reference and mesh sensitivity of the three-dimensional solver

Figure S3 shows the free-space check:  $T_B$  stays within  $[0.993, 1.003]$  while  $F_{p,B}$  oscillates between 0.93 and 1.13, which yields the unphysical  $\eta_B = 1.08$  at 748 nm and bounds the normalisation uncertainty of  $F_p$  at  $\pm 13\%$  (3% at 610 nm). Figure S4 shows the mesh sensitivity of case A: the radiative peak position (616(4) nm) and the efficiency at it are stable, whereas the peak heights depend on the sub-cell position of the source. Ratios between cases computed with the same mesh and source position are unaffected.

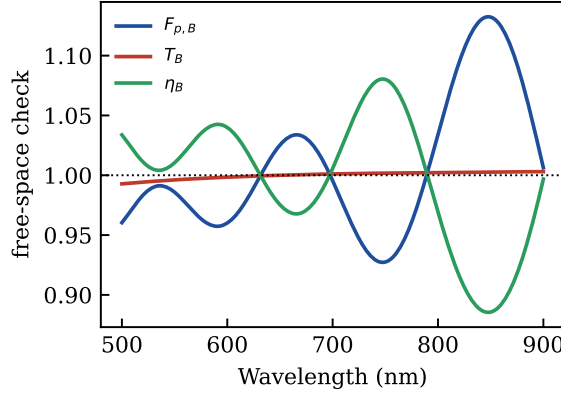


Figure S3: Free-space reference B of the three-dimensional solver.

## S6. Body-of-revolution solver: method and benchmarks

Cylindrical-coordinate FDTD (Meep [2]) with field dependence  $e^{im\varphi}$ ; axial dipole:  $E_z$  source,  $m = 0$ ; transverse dipole:  $E_r$  source,  $m = 1$ .  $T$  is the ratio of the fluxes through a closed surface (top and bottom discs and a cylindrical side wall) around the particle with and without it.  $F_p = (P_{\text{rad}} + P_{\text{abs}})/P_0$ , where  $P_{\text{abs}}$  is the net inflow through a second closed surface that encloses only the particle, its top disc lying at mid-gap. The local density of states at the source cell gives the same order of magnitude but is spiky near the metal and is used only as a cross-check. The band 490 nm to 900 nm is assembled from three runs with pulses covering 460 to 610, 540 to 760 and 630 nm to 990 nm, of which only 490 to 570, 570 to 700 and 700 nm to 900 nm are kept: a

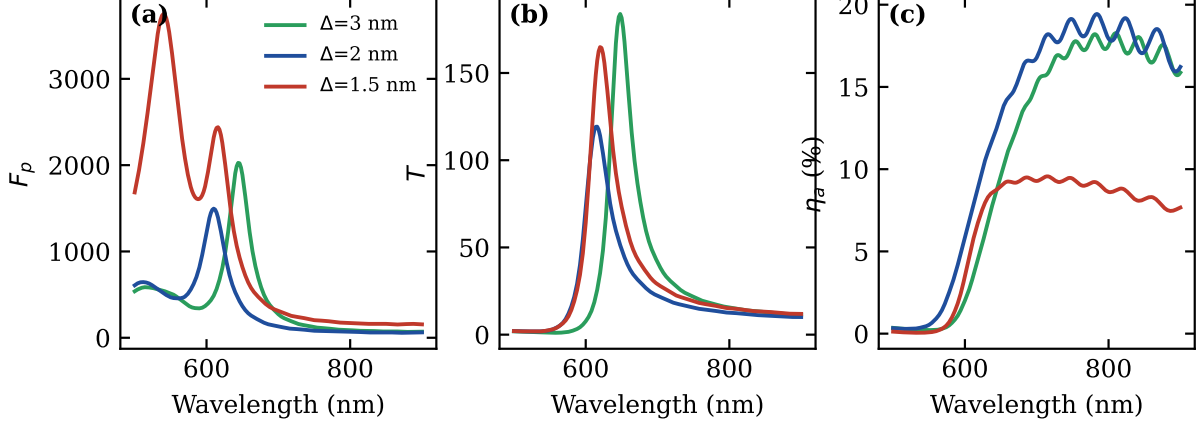


Figure S4: Mesh sensitivity of case A in the three-dimensional solver: (a)  $F_p$ , (b)  $T$ , (c)  $\eta_a$  for override meshes of 3, 2 and 1.5 nm.

broadband single run produced errors up to 37% at the band edges, where the source spectrum is weak. Run time 30  $\mu\text{m}/c$  after the source (identical  $F_p$  to 60); PML thickness 0.15  $\mu\text{m}$ , with a 0.3  $\mu\text{m}$  check at the finest grid. To avoid long-lived numerical ringing, the Lorentz-pole damping in the gold fit is bounded below by 0.2  $\mu\text{m}^{-1}$  (same fit accuracy).

Benchmark against the exact Mie solution for a dielectric sphere ( $\varepsilon = 4$ ,  $R = 15.874$  nm, radial dipole at 5 nm): a grid of 2 nm reproduces  $F_p$  within 1% in the band centres; a 4 nm grid gives a uniform 4% to 5% overestimate with a maximum deviation of 12%. For gold (radial dipole, 5 nm gap) at a 2 nm grid,  $T$  deviates from Mie by about 10% while the power-balance  $F_p$  is about three times too large; at a 20 nm gap the  $F_p$  error drops to 24% and  $T$  agrees within 5%. The  $F_p$  error is therefore a near-field discretisation error of the staircased metal surface. Grids of 2, 1.33 and 1 nm with linear extrapolation to zero spacing are summarised in Table S6.

Table S6: Body-of-revolution solver against exact Mie theory (gold sphere, radial dipole, 5 nm gap) and grid convergence of the nanorod (axial dipole, 5 nm gap). Errors are medians over 490 nm to 900 nm.

| grid (nm)    | sphere: $ \Delta F_p $ | sphere: $ \Delta T $ | rod: $T_{\max} (\lambda_T)$ | rod: NA 0.9 fraction |
|--------------|------------------------|----------------------|-----------------------------|----------------------|
| 2.00         | 215%                   | 10.2%                | 149 (631 nm)                | 0.181                |
| 1.33         | 108%                   | 8.4%                 | 134 (617 nm)                | 0.181                |
| 1.00         | 93%                    | 7.2%                 | 125 (611 nm)                | 0.180                |
| extrapolated | 48%                    | 4.4%                 | —                           | —                    |

PML thickness 0.15 versus 0.3  $\mu\text{m}$  at the 1 nm grid changes  $T$  by 0.1% (median) but  $F_p$  by 9%. The radiative quantities are therefore converged to the level quoted in the main text, whereas  $F_p$  in a 5 nm gap is not converged at grids of 1 nm with a staircased surface and is not used; all  $F_p$  values in the main text are BEM results. The transverse ( $m = \pm 1$ ) configuration with a single  $E_r$  source on the axis gave unphysical (negative) total rates and is likewise not used; all transverse results in the main text are BEM results. All runs were executed on GitHub-hosted runners (workflow [.github/workflows/meep-runs.yml](https://github.com/workflows/meep-runs.yml)); raw outputs are in `isi/results/ci`.

## References

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